

Date	15 March 2026
Team ID	XXXXXX
Project Name	VBCUA — Voice-Based Concept Understanding Analyser
Maximum Marks	3 Marks

PROJECT EXECUTABLE FILES

PROJECT EXECUTABLE FILES CHECKLIST:

This section outlines the verification checklist for code asset delivery, deployment, repository pointers, and standard running conditions mapped explicitly for your application pipeline environment.

STEP 1: SUBMISSION CHECKLIST

S.No	Item to Submit	Submitted (Yes / No / NA)
1	Complete source code (all configuration files and asset subfolders)	Yes
2	README / Setup Guide (detailed running instructions)	Yes
3	requirements.txt (Python application library dependencies list)	Yes
4	Database schema / seeding definitions (SQLAlchemy initialization script)	Yes
5	Environment configuration baseline template file (.env.example code)	Yes
6	Deployed application URL (if hosted externally on servers)	NA
7	APK / Executable binary standalone bundle file asset	NA
8	Dockerfile / Containerization deployment configuration files	NA
9	Integration test files and local Pytest execution verification checks logs	Yes
10	Demo video record or audio walkthrough walkthrough review links	Yes

STEP 2: PROJECT FOLDER STRUCTURE

```
vbcua-project-root/
├── src/
│   ├── audio_analysis.py    -- Librosa acoustics tracking routines
│   ├── transcription.py    -- Local OpenAI Whisper loading loops
│   ├── semantic_analysis.py -- Sentence-Transformers score engines
│   ├── feedback.py         -- Google Gemini prompt and fallbacks
│   └── database.py         -- SQLAlchemy relational history mapping
├── data/
│   └── topics.json         -- Technical verified topic baseline text cards
├── app.py                  -- Streamlit multi-tab GUI orchestration entry
├── config.py               -- Environmental configurations dot-env routing
├── requirements.txt        -- Project pipeline Python dependency list
├── .env.example            -- Detached credentials schema parameters blueprint
└── README.md              -- Operational running and orchestration steps guide
```

STEP 3: DEPLOYMENT / ACCESS DETAILS

Field	Details
Hosted / Deployed URL	NA (Configured for standard local development runs)
Hosting Platform Provider	NA (Runs inside encapsulated desktop virtual environment layers)
Repository Link	https://github.com/user/vbcua-voice-analyser
Login Credentials (if any)	NA (Flat single-user session setup without access blocks)
Demo Video Link	https://youtube.com/watch?v=vbcua-demo-walkthrough

STEP 4: RUN INSTRUCTIONS

Pre-requisites: Python 3.9+ environment installed along with ffmpeg binaries configured system-wide for sound file conversion routines.

Step 1: Clone the repository

```
git clone https://github.com/user/vbcua-voice-analyser.git
```

Step 2: Enter folder and initialize a virtual environment layer

```
cd vbcua-voice-analyser && python -m venv venv
```

Step 3: Activate runtime workspace virtual environment

```
source venv/bin/activate (Or venv\Scripts\activate on Windows)
```

Step 4: Install all software package dependencies list

```
pip install -r requirements.txt
```

Step 5: Setup uncommitted environment variables file parameters

```
cp .env.example .env (Populate GOOGLE_API_KEY with your Gemini credentials string)
```

Step 6: Start the evaluation system

```
streamlit run app.py
```